

José Manuel Robles-Aguilar

M.SC. IN STATISTICAL PHYSICS · ARTIFICIAL INTELLIGENCE · COMPUTATIONAL BIOPHYSICS

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Education

Cinvestav

Mexico City, MX

M.SC. IN PHYSICS

01 Sep 2023 - 16 Dec 2025

- **Thesis:** AI-driven Prediction of Circular Dichroism Spectra of Proteins.
- **Advisor:** Dr. Damián Jacinto-Méndez
- **Focus:** Integration of Deep Bayesian Neural Networks with the theoretical KCD method.
- **GPA:** 9.1/10

National Polytechnic Institute (IPN)

Mexico City, MX

B.SC. IN PHYSICS AND MATHEMATICS

01 Sep 2017 - 01 Jul 2022

- **Focus:** Stochastic Processes and Materials Science.
- **GPA:** 8.64/10

Research Interests

Computational Biophysics, Molecular Dynamics (MD), AI-driven Scientific Discovery, Peptide/Protein Design, Lipid-based Nanocarriers (LNCs), Viral Fusion, G-Protein-Coupled Receptors (GPCRs), Membrane Proteins.

Research Experience

Graduate Researcher

Sep 2023 - Dec 2025

DEPT OF PHYSICS, CINVESTAV, MEXICO CITY, MX

- Developed a hybrid AI-Physics framework to predict protein optical activity, outperforming state-of-the-art methods by 40%.
- Modeled electromagnetic interactions in proteins using the DeVoe polarizability model.
- Implemented Deep Bayesian Neural Networks to quantify uncertainty in spectral predictions, addressing the "black box" problem in AI-driven biophysics.
- Analyzed complex biological datasets to correlate structural motifs (Alpha-helix/Beta-sheet/Random Coils/Others) with spectral outputs.

Research Intern - Soft Matter Laboratory

May 2024 - Jul 2024

CINVESTAV, MEXICO CITY, MX

- Investigated Brownian motion of nanoparticles within lipid vesicles.
- Designed a computer vision pipeline for particle tracking and trajectory analysis.
- Calibrated high-precision fluorescence microscopy equipment and captured imaging data across varying focal heights.
- Contributed to the understanding of diffusion processes in biological soft matter systems.

Teaching Assistant

Sep 2021 - Mar 2022

CENTER FOR COMPUTING RESEARCH (CIC), MEXICO CITY, MX

- Translated complex technical course material into accessible educational animations using Adobe Illustrator and After Effects.
- Taught and prepared materials for courses on LaTeX and technical documentation in English and Spanish.

Research Fellow (BEIFI Grant)

2018 - 2022

SUPERIOR SCHOOL OF PHYSICS AND MATHEMATICS (ESFM-IPN), MEXICO CITY, MX

- Conducted experimental research and computational simulations in Materials Science and Stochastic Processes.

Projects

Drug-Protein Binding Affinity Prediction using GNNs

Feb 2026 - Present

PERSONAL PORTFOLIO

- Developing a Graph Neural Network (GNN) architecture to predict the binding affinity between drug molecules and target proteins.
- Modeling molecular structures as graphs to capture spatial and topological features essential for accurate affinity scoring.
- Utilizing GPU resources for efficient training on large-scale bio-molecular datasets.

MD simulations of a mutation in the human H2 receptor (hH2R)

Mar 2026 - Present

DEPARTMENT OF PHYSIOLOGY, BIOPHYSICS AND NEUROSCIENCE, CINVESTAV

- Investigating the mutation N217D in the human H2 receptor (hH2R) using Molecular Dynamics (MD) simulations, in collaboration with the PhD student Miroslava Carrasco.

Conferences and Schools

13th Symposium on Structural Proteomics

Milan, Italy

POSTER PRESENTER

Oct 2025

- Presented research on AI-driven CD spectra prediction to an international audience of computational biophysicists and structural biologists.

Physics and Life Days Conference

Paris, France

POSTER PRESENTER

Oct 2025

- Presented the Knowledge-based Circular Dichroism (KCD) method to an international audience of physicists.

AI+Science Summer School

Paris, France

ATTENDEE

Jul 2025

- Selected to attend intensive training on the application of Artificial Intelligence in scientific discovery.

QMUL-Durham-Leeds Research School

Durham, England

ATTENDEE

Jan 2022

- Collaborated with an international team to develop and defend a research proposal addressing Sustainable Development Goals (SDG).

Technical Skills

Machine Learning & AI: Deep Learning, Bayesian Neural Networks (BNNs), Graph Neural Networks (GNNs), Computer Vision, Regression, Clustering, XGBoost, LightGBM, GenAI, Prompt Engineering, Predictive Modeling, Hyperparameter Optimization.

Programming & Tools: Python (PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Optuna, Seaborn, GROMACS, MDAnalysis, NGLView), Git, Linux/Unix, Bash Scripting, CUDA/GPU Acceleration, SQL, Excel, MATLAB, Adobe Illustrator, After Effects, Power BI, LaTeX.

Scientific Computing: Bioinformatics, Molecular Dynamics, Proteomics, ChimeraX, Statistical Mechanics, Monte Carlo Simulations, Numerical Methods, Quantum Computing, Exploratory Data Analysis, Data Science.

Experimental: Fluorescence Microscopy, CD Spectroscopy, XRD, AFM, EM, UV-Vis, Raman.

Mathematics: Bayesian Statistics, Linear Algebra, Calculus, Stochastic Processes, Game Theory, Non-equilibrium Statistical Mechanics.

Languages: Spanish (Native), English (C1, IELTS: 7.5), French (B1), German (A1).

References

- Dr. Damián Jacinto-Méndez**, Cinvestav, Department of Physics. **email:** damian.jacinto@cinvestav.mx
- Dr. José Matías Alvarado-Mentado**, Cinvestav, Dep of Computer Science. **email:** matias@cs.cinvestav.mx